CoLoRe installation instructions for NERSC

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1 Create conda environment

Create an empty conda environment and install the following C libraries from the `conda-forge` channel: `libconfig, libsharp, cfitsio, healpix`. You can do so with the following commands:

```
conda create -n colore_env
conda activate colore_env
conda install -c conda-forge libconfig
conda install -c conda-forge libsharp
conda install -c conda-forge cfitsio
conda install -c conda-forge healpix
```

2 Environmental variables

Make sure that the libraries used to compile CoLoRe are in `LD_LIBRARY_PATH` so they can be found at runtime. Open your `.bashrc` file and make the directories point to your new environment where the libraries have been installed. You can do so by copying the following lines into your `.bashrc` file:

```
export FFTW_DIR=/opt/cray/pe/fftw/3.3.10.11/x86_milan
export CFITSIO_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export HEALPIX_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export LIBSHARP_DIR=/global/common/software/desi/user/.conda/envs/colore_env
export LIBCONFIG_DIR=/global/common/software/desi/user/.conda/envs/colore_env

export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/lib64:${FFTW_DIR}/lib:${CFITSIO_DIR}/lib:${LIBSHARP_DIR}/a
↪  uto/lib:${LIBCONFIG_DIR}/lib
```

3 Modify CoLoRe's makefile

Clone the desired branch of the CoLoRe GitHub repo into your home directory:

```
git clone https://github.com/damonge/CoLoRe/tree/branch
```

Inside the repo folder `./CoLoRe`, modify the `Makefile` to use the environmental variables defined earlier in the `.bashrc` file. A few other variables have to be changed since NERSC has continued to update its modules and libraries. For example, the compiler options need the `-fpermissive` tag to allow for some differences between the compiler's NERSC and CoLoRe versions. Below you can find an example of the the modified `Makefile`.

```
########## User-definable stuff ##########
###Compiler and compilation options
COMP_SER = cc
COMP_MPI = cc
OPTIONS = -Wall -O3 -std=c99 -fpermissive

### Behavioural flags
#Use double precision integer (enable in general)
DEFINEFLAGS += -D_LONGIDS
```

```
#Use normalized bias model
DEFINEFLAGS += -D_BIAS_MODEL_2
#Use linear bias model
#DEFINEFLAGS += -D_BIAS_MODEL_3
#Use new lensing method
#DEFINEFLAGS += -D_USE_FAST_LENSING
#Generate debug help. Only useful for development
DEFINEFLAGS += -D_DEBUG
#Use double precision floating point? Set to "yes" or "no"
USE_SINGLE_PRECISION = yes
#Add random perturbations to kappa from redshifts outside the box
#ADD_EXTRA_KAPPA = no
#Compile with HDF5 capability? Set to "yes" or "no"
USE_HDF5 = no
#Use OMP parallelization? Set to "yes" or "no"
USE_OMP = yes
#Use MPI parallelization? Set to "yes" or "no"
USE_MPI = yes

###Path to libraries and headers
###If two or more of the dependencies reside in the same paths, only
###one instance is necessary.
#GSL
#GSL_INC = -I/add/path
#GSL_LIB = -L/add/path
GSL_INC = -I$/usr/include
GSL_LIB = -L/usr/lib64 -lgsl -lgslcblas

#FFTW
FFTW_INC = -I${FFTW_DIR}/include
FFTW_LIB = -L${FFTW_DIR}/lib

#cfitsio
FITS_INC = -I${CFITSIO_DIR}/include
FITS_LIB = -L${CFITSIO_DIR}/lib -lcfitsio

#cfitsio
HDF5_INC =
HDF5_LIB =

#libconfig
CONF_INC = -I${LIBCONFIG_DIR}/include
CONF_LIB = -L${LIBCONFIG_DIR}/lib

#healpix
HPIX_INC = -I${HEALPIX_DIR}/include
HPIX_LIB = -L${HEALPIX_DIR}/lib

#libsharp
SHT_INC = -I${LIBSHARP_DIR}/include
SHT_LIB = -L${LIBSHARP_DIR}/lib
```

This last part of the file remains the same. Notice how this `Makefile` uses the normalized bias model. If you want to use any other bias model such as the linear bias one, comment the normalized bias model lines and uncomment the linear bias ones.

4 Compile CoLoRe

Before compiling CoLoRe, you need to load the NERSC provided version of the FFTW Library so that the compiler can find the FFTW headers and libraries. You can do so by running:

```
module load cray-fftw
```

Now everything is ready to compile the CoLoRe code. Go to the CoLoRe repo folder and use the `make` command inside it. To make sure it has compiled correctly, no error should appear as the last messages of the compilation.

Remember

Everytime CoLoRe is run from a script, the conda module has to be loaded and the environment has to be activated. You can just copy the following lines into your script.

```
module load conda
conda activate colore_env
```

Also, if you want to change your bias model at any point, you must modify the `Makefile` again, uncommented the bias model you want to use and commenting the rest. Then, you should clean temporary files, compiled binary objects, and executables and compile again typing:

```
make clean all
make
```